Section 13.3: Alternating Current

Section 13.3 Questions, page 598

1. No, electrons move back and forth about the same spot at every point along the conductors from the power plant to my home to provide the electrical energy.

2. Yes, the voltage is proportional to the current. In alternating current electricity, as the voltage increases, the conventional current in the wire increases in the positive direction and as the voltage decreases, the conventional current in the wire decreases. This shows that the voltage is proportional to the current.

3. (a) Yes, if the frequency of the alternating current were reduced to 2 Hz it would be noticeable, particularly in the operation of light bulbs. A frequency of 2 Hz means that the current goes in a positive direction, reverses, and then goes in a negative direction two times each second. Any light bulb connected to the circuit would become brighter and dimmer twice each second, which would be very noticeable.

(b) The frequency may have been increased to reduce the flicker in electric lights.

4. A 120 V plug has three prongs and supplies appliances with an effective voltage of 120 V.

A 240 V plug has four prongs and supplies appliances with an effective voltage of 240 V.

A 240 V plug can also supply an appliance with an effective voltage of 120 V for parts of the appliance that do not require 240 V.

5. There must be an adapter that is connected between the laptop and the wall outlet that converts the alternating current into direct current.

6. A fuse is a device that is placed in series in a circuit. The fuse melts and opens the circuit if the current exceeds some maximum value at which the fuse is rated. A fuse must be replaced to close the circuit again.

A circuit breaker functions in a similar way as a fuse, except it does not melt. A circuit breaker is made with a bimetallic strip that bends to open the circuit if there is too much current in the wire. A circuit breaker can also be reset to close the circuit again.

A GFCI is similar to a circuit breaker. A GFCI is more sensitive to very small changes in current and opens the circuit much more quickly than a circuit breaker.

An AFCI is similar to a GFCI. It acts quickly to shut off a circuit when an arc along the circuit is detected.

7. The difference between the voltage used in household circuits in North America and Europe is primarily due to historical choice. Once a voltage was adopted for use in an area, the distribution systems and electric machines and appliances were designed for that specific voltage, so it would be very impractical to change the voltage used. There are very few differences when using the different power systems so one system is not better than another.